



Agentic NLP for Thematic Analysis of Fed Speeches: Yield Impacts under Jerome Powell (2018-2025)

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MOTIVATION, DATA & OBJECTIVES

THE PROBLEM & GAP



Fed communications guide trillions. Under Powell, narrative shifted.

Traditional NLP (BoW, LDA) lacks granularity & reasoning capabilities.

OBJECTIVES



✓ **Technical:** Evaluate Agentic AI (LangGraph/LLaMA) for structured economic text mining.

✓ **Financial:** Assess if narrative emphasis predicts abnormal 10-year Treasury yield changes.

DATA COLLECTION



Source: 1,200+ speeches from Federal Reserve archive (2008-2025).

Tooling: Selenium-based scraper, PostgreSQL storage (raw & structured).

METHODOLOGY: THE AGENTIC NLP PIPELINE (LangGraph)

Input: Raw Fed Speeches
(Unstructured Text)

LangGraph Orchestration Loop (State Management & Retries)

Agent 1: Theme Classifier (LLaMA 3.1 & Pydantic)

- Splits speeches into paragraphs.
- Classifies as Inflation, Employment, or Other with strict type-checking.

Agent 2: Emphasis Scorer (Aggregation Agent)

Aggregates paragraph tags into compositional score vector (Sums to 1) for each speech.



Key Advantages: Stateful execution, error handling, parallel processing, superior to monolithic prompts.

Output:
Structured JSON
(Theme Weights per Speech)
→ PostgreSQL

EVENT STUDY FRAMEWORK

$$CAY_t = \beta_0 + \beta_1 I_t + \beta_2 E_t + \beta_4 FFR_t + \epsilon_t$$

Metric: Cumulative Abnormal Yield (CAY).

Control: Fed Funds Rate (FFR) to isolate narrative impact.

RESULTS & CONCLUSION

VISUAL ANALYSIS (THEMATIC EVOLUTION)

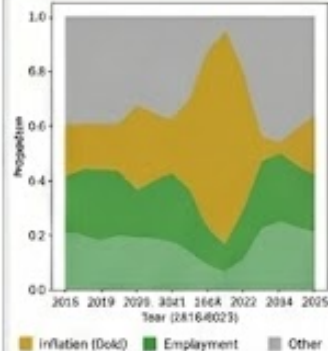


Figure 1. Inflation Focus Spikes in 2022

REGRESSION RESULTS (KEY FINDINGS)

Variable	Coefficient	P-Value
Inflation Emphasis	-0.0145	0.171
Employment Emphasis	0.0037	0.740
Fed Funds Rate	-0.0028	0.003

Table 1. Impact on 10Y Treasury Yields

Interpretation: Markets react significantly to the rate decision itself (FFR), not the thematic proportion of the accompanying speech.

CONCLUSION & FUTURE WORK



Conclusion: Agentic NLP successfully structured text at scale, offering a reproducible mining foundation.

Future Work: Incorporate Sentiment Analysis (Tone) and hierarchical paragraph weighting.